Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/04/2020
PI Name:	Reschke, Millard F Ph.D.		
Project Title:	Orion Crew Performance and Safety Immediately Following Postflight Water Landings: Emergency Egress		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	millard.f.reschke@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-7210
Organization Name:	NASA Johnson Space Center		
PI Address 1:	2101 NASA Pkwy # ONE, SK272		
PI Address 2:	Neuroscience Laboratories		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO 80JSC017N0001-Crew Health and Performance (FLAGSHIP1, OMNIBUS). Appendix A-Flagship1, Appendix B-Omnibus
Start Date:	11/01/2018	**End Date:**	07/26/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/26/2020 due to augmentation and PI change to Dr. Brian Peters, per HHC/HRP (Ed., 8/5/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bloomberg, Jacob Ph.D. (NASA Johnson Space Center) Mulavara, Ajitkumar Ph.D. (NASA Johnson Space Center) Caldwell, Erin M.S. (KBR/NASA Johnson Space Center) Clement, Gilles Ph.D. (KBR/NASA Johnson Space Center) Peters, Brian Ph.D. (KBR/NASA Johnson Space Center) Rosenberg, Marissa Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The decision to return to water landings in the Orion capsule presents significant safety issues that must be addressed. First, based on current models, it is anticipated that stable 2 capsule attitude upon splash down will occur more that 51% of the time and remain in that position for up to 20 min. Second, currently 100% of returning International Space Station (ISS) crewmembers (both Russian and US Orbital Segment (USOS)) have landing sickness, the severity of which is expected to increase due to wave motion coupled with exposure to a closed visual environment (cannot see a horizon). Third, strength, while compromised, is currently not believed to be an issue for emergency egress in long duration crewmembers (20+ days). However, crewmembers are vestibular and sensorimotor challenged to the point where coordinated walking, rising from a seat, or standing is difficult. Fourth, 100% of returning Soyuz crews when questioned about an unaided emergency egress from the Orion in the water have said that it will be catastrophic. Based on current neurosensory studies (Functional Mobility, Functional Task Test, and the Field Test) the goals of this study are four-fold: First, use our laboratories' 6 degree-of-freedom platform (6-dof) to develop a motion based simulation of potential sea states expected following an Orion landing. Second, investigate the effect of a moving platform on functional performance in a normative population. Third, develop and test minimal noninvasive countermeasures to counteract the effect of a moving platform. Fourth, make the 6-dof platform simulation available for use by other NASA divisions interested in testing Orion landing concepts/procedures (e.g., currently, demonstrations and tests of Orion escape involve a 4 hr roundtrip to the Gulf of Mexico (GoM), transport of the capsule, etc.). Proposed Studies: (a) Effect of a Moving Platform on Performance: There is considerable evidence that exposure to a moving platform can have significant effects on several performance factors essential for safety and egress from the platform apart from the difficulties astronauts experience after landing. General effects are seen when subjects are introduced to a moving platform: reduced motivation, increased reaction times, fatigue, modified motor skills, and significant balance problems (1). We intend to investigate the effects of platform motion (using Orion 3-d accelerations prerecorded during GoM trials) to those observed from our Field Test postflight to provide a baseline of effects expected from a postflight, in the water, Orion environment. (b) Evaluate Orion Egress with and without countermeasures: during everyday life, we develop through voluntary movement a neural image (efference copy) of expected upcoming afferent movement patterns (reafference). On a moving platform (novel environment), the reafference received in the brain from vestibular and proprioceptor systems during movement will not match the brain's efference copy. That is, the movement we normally make will not match the movement required. Without vision, it is difficult to develop a new set of movements corresponding to the environmental demands. Inside the Orion capsule, the crews have no visual signals that correspond to the motion (worst possible environment). A countermeasure consisting of a Malcolm Horizon (MH) device can alleviate this problem. The MH provides an attitude indicator, displaying roll and pitch of the capsule (a continuous 360-degree laser line visible in the capsule) relative to the gravitational vertical. It uses the visual system to subconsciously maintain continuous control and awareness of personal attitude (2). Other simple countermeasures can be evaluated: hand-holds appropriately placed, crawling as opposed to upright locomotion, etc. References: 1. Wertheim AH. Working in a moving environment. Ergonomics, 41(12): 1845-58. PubMed PMID:9857842. 1998. 2. Malcolm R. The Malcolm Horizon. Proceedings of the Peripheral Vision Horizon Display. NASA Edwards, CA, (pp. 12-40). 1983. NOTE: Continued by "Orion Crew Performance and Safety Immediately Following Postflight Water Landings: Emergency Egress (PI Peters)" with new Principal Investigator Dr. Brian Peters.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal effort set out to develop a test that could be performed on returning crewmembers to better quantify astronauts' ability to egress the capsule after a water landing. A secondary goal was to compare the performance between a condition during which an Earth-fixed visual reference was projected onto the walls of the enclosure and another without the visual cue. An Earth-fixed visual reference has applications for other motion platforms, including those on air, land, or sea.
	Hardware Development Wave Motion Creation: A six degree-of-freedom motion platform (Moog, Inc., East Aurora, NY), was used to generate the wave motion simulations used for this investigation. Data from Inertial Measurement Units (IMUs), placed inside the Orion capsule during egress tests performed in the Gulf of Mexico, were used to generate the simulated waves. Analysis of the time series was performed in Matlab (MathWorks, Natick, MA). The pitch and roll time series data were converted into the frequency domain using a Fast Fourier Transform (FFT). The result provided information regarding the range of frequencies, the relative amplitudes and the phase of each component sine wave present in the data. Independent sum-of-sines equations were derived from this information for the pitch and roll movements. Each equation was comprised of 3 sine waves. The relative frequencies and associated phase for these sine waves were chosen to stay within the range of the frequencies from the measured values. To prevent having the simulated wave repeat itself within a short time period, the chosen frequencies were not easy multiples of one another. The equations were used to drive the pitch and roll axes of the motion platform. Using a movement profile created from a combination of pitch and roll movements would just result in random tilts of the floor rather than the rolling motion of a wave. To make the wave simulation more realistic, heave (i.e., vertical) motion was added. Empirical data were not used to generate this motion. Instead, the pitch and roll equations were used to create the time series data from which the heave equation was derived. The roll and pitch timeseries data were added together to create a single wave. An Functional Task Test (FTT) was again used to convert this combined wave into the frequency domain. The phase of the component sine was offset by 45 degrees for each of the waves at the same individual frequencies used for the pitch and roll waves. The relative amplitudes for the sine wave comprising the heave signal were consistent with the FFT result, but the baseline amplitude was determined through empirical trial and error. The custom drive software (LabView, National Instrument, Austin TX) has the ability to apply a simple multiplier to the drive equations so the process of creating a LOW and HIGH wave state was accomplished by multiplying the LOW wave amplitudes by two. The result was an ability to apply the same wave profiles for each wave state condition, differing only in overall amplitude. Visual Surround: The visual environment will be a significant sensory challenge while in the capsule. It is not the visible objects in the environment, but rather the fact that the visual environment will not stay fixed relative to gravity that creates the problem. To simulate these visual conditions, an enclosure was affixed to the surface of the motion platform.

The three-walled “room” was constructed from extruded aluminum (80/20 Inc., Columbia City, IN) and wallpaper-covered foam core. The walls were perpendicular to the floor, but the side walls were not perpendicular to the front wall. Testing the spatial orientation countermeasure required that a laser line be projected into the enclosure from the open end. To create flexibility in the positioning of the laser projectors, while still assuring that the line be visible on the side walls, each wall was flared by approximately 5 degrees. Because the center of gravity of the Orion capsule is not at its geometrical center, the capsule does not float with the floor perpendicular to the gravitational vector. For the Orion capsule the nominal orientation of the floor, in calm water, will be 3 degrees from horizontal. In some landing configurations it can be greater than 25 degrees. The commercial crew capsules will also have canted floors, but the degree to which they slope was not made available. To replicate this feature of the capsules, the resting angle of the enclosure for this investigation was set to 10 degrees.

Artificial Horizon: The spatial orientation countermeasure was simply a horizontally presented laser line. Two self-leveling laser projectors (Huepar Box-1G) were used to generate the green laser line. The projectors were mounted outside of the enclosure, so they were not subjected to the movements of the motion platform. Because the subject stood between the projector and the wall upon which the line appeared, the line was interrupted by a “shadow” cast by the subject. Two laser projectors, horizontally separated by 52 in, were used to minimize the break in the otherwise continuous line. The line obviously moved with respect to the walls of the enclosure during the wave conditions, but height of the line was approximately 5 ft above the floor if measured on the front wall at the midline of the non-moving enclosure.

Subjects: Thirty-two subjects were recruited through the NASA Test Subject Support Facility. Subjects passed a NASA-modified Air Force Class III physical prior to enrollment. Signed informed consent for NASA Institutional Review Board-approved procedures was obtained from the subjects prior to any testing.

Procedures

Subjects participated in a single data collection session. They changed into lab-provided shoes to keep the footwear consistent between subjects and were also outfitted with a fall protection harness and instrumented with data collection hardware. After an explanation of the tasks and a demonstration, the subjects were allowed to practice the tasks until they felt comfortable with the requirements and sequence of the activities. For the practice trials, the motion platform was moved to make the floor of the enclosure parallel with the ground and held static. Data were collected during four trials; one for each of four conditions. The order of the conditions was randomized for each subject. The four conditions were:

STATIC – base was held in the neutral position (enclosure rolled 10 degrees)

LOW – wave motion using equations provided above

HIGH – wave motion at double the amplitude of the LOW condition

HIGH w/ line – repeat of the HIGH condition, but with the projected laser line

During each trial the subject performed a sequence of tasks designed to produce movement patterns that would be present during capsule egress activities. The tasks involved the manipulation of a “rope” ladder and a weighted bag. Subjects started the trial from a prone position. The trial started at the sound of a tone, after which the subject stood and pressed a switch in the center of the front wall at a height of approximately 5 feet. Pressing the switch started a 30 second countdown timer. Subjects were instructed to stand quietly, facing the front wall, during this thirty second period. Another tone sounded at the end of this quiet stance period and the subject again pressed the wall-mounted switch. A “rope” ladder was then deployed. The top of the ladder was permanently attached to the front wall. When deployed, the bottom of the ladder was attached, using two carabiner-type clips, to floor-mounted eyebolts. In the initial position, the clips at the bottom of the ladder hung just above the top of the ladder. To deploy the ladder, subjects had to lift the clips off of the hangars, attach them to the eyebolts and then cinch up each side of the ladder by pulling on the ends of the two nylon straps. A spring-loaded cam buckle held the strap taut after it was pulled. After again pressing the wall switch at the end of the ladder deployment task, the subject lifted the weighted bag and hung it on a hook mounted on the front wall. After hanging the bag, the deployment tasks were complete, and the subject again pressed the wall switch. The trial continued as the subjected returned the ladder and bag to their original starting positions, starting with the ladder. The wall switch was pressed after each step.

Data Collection and Analysis: The primary dependent variable for this investigation was completion time. A custom software program (LabView, National Instruments, Austin, TX) was used to control the protocol execution and record the times. Each time the wall switch was pressed the elapsed time since the beginning of the trial was logged. It wasn’t necessary to have each phase of the task timed but forcing the subjects to push the switch between each required them to make additional postural adjustments during the execution of the activity. The time required to complete the deployment and the total trial time were the focus of the analysis. Emerald IMUs (ADPM, Inc., Portland, OR) recorded kinematic data from the head and torso during the trials. When the focus is a comparison of performance between conditions, it is not rational to analyze overall subject movement when the underlying motion of the platform differs. Similarly, the goal of the investigations was not to compare one strategy from another between conditions. An analysis is not necessary to conclude that a subject moves more when the support surface is moving than when it does not and it isn’t necessary to look at movement difference when one subject might bend at the waist and another might choose to kneel to accomplish the same task. Therefore, the analysis of the kinematic data was restricted to the quiet stance portion of the HIGH and HIGH w/ line conditions. Because the start of the platform motion and the start of the data collection trial were not synchronized, it was possible that the specific wave motion experienced could differ between conditions. Care was taken to assure that only portions of the quiet stance that had the same wave motions were compared. A cross-correlation calculation was done using data from a third IMU affixed to the floor during each trial. The peak correlation provided an indication of the time delay between the start of the quiet stance periods with respect to the underlying wave motion. This delay was accounted for and only data from the portion of the quiet stance period where the wave motion was identical were compared. A resultant acceleration vector was calculated from the triaxial accelerometers in the torso mounted IMUs. A root mean square (RMS) analysis of the magnitude of this vector provided an indication of the subjects’ linear movements. Similarly, the RMS of the combined pitch and roll movements of the subject was determined to quantify the rotational movements. All analyses were performed using custom Matlab (MathWorks, Natick, MA) programs.

Statistical Analysis: A linear regression model was constructed to quantify the association between each split time and

Task Progress:

condition (static [S], low wave [L], high wave [H], high wave with a horizontal line [HL]) and order (1,2,3,4). All model parameters were estimated using generalized estimating equations with an independence correlation structure to account for multiple measurements per subject. The Wald test was used to evaluate model fit using 4 different models (main effects only, interactions, factor codings). The final model included main effects of condition and order as well as their interaction. Linear combinations of coe?icients were computed to summarize main effects (e.g., S when order = 1), within order comparisons (e.g., HL vs S when order = 1), and within condition comparisons (e.g., 1 vs 2 = 2 vs 3 = 3 vs 4 for S). A similar modeling approach was taken to summarize the association between total time and condition/order. The movement analysis mimicked the approach which was described above. Regardless of the outcome (roll/pitch and mean acceleration using torso measurements), the final models regressed movement onto main effects of horizontal line (yes/no) and order (first, second). Linear combinations of coefficients were computed to summarize main effects (e.g., horizontal line = No, order = first) and horizontal line differences (Yes-No).

RESULTS

Completion time results were analyzed using data from 30 of the 32 subjects. Data from the first two subjects were eliminated because the shadows on the walls of the enclosure, created by the room lights, may have provided them with undesirable cues. As expected, the shortest completion times occur when the platform is not moving, and the highest values occur during the HIGH wave conditions.

A learning effect contributed significantly to the variability of the completion time data ($p < 0.000$). This condition was the most balanced of the conditions and not subject the confounds of platform motion that effect the other three conditions. The test order for each subject was randomized but doing so did not result in the desired balance in the test design. In an effort to determine if the presentation of an Earth-fixed horizontal line on the walls of the enclosure improved stability, a comparison was also made for kinematic parameters gathered during the quiet standing period of the HIGH and HIGH w/ line conditions. Because of the requirement to make comparisons between sections of the data where the same wave motion was presented during each condition, only 19 of the 30 subjects had enough data overlap between conditions to be reasonably included in this analysis. The group means appear to indicate that the presence of the line had a negative effect on linear movements of the torso. Indeed, statistical analysis indicates a trend toward this conclusion. This is not the case for the rotation parameter in 4B. Three subjects in the HIGH w/ line condition (4A) seemed to be elevating the mean for the group prompting a closer look at these subjects. For all three, the HIGH w/ line condition occurred before the HIGH condition. Like the completion time data, a learning effect was present in kinematic parameters extracted from the quiet stance period ($p = 0.012$ & $p = 0.015$ for the translation and rotation parameters, respectively). Translation and rotation measures both show a reduction in the magnitude of the subjects' movements during their second exposure to the higher waves, regardless of the absence or presence of the visual cue. Subject reporting contradicts these empirical results. When asked after completion of all of the conditions whether the presence of the horizontal line helped them stabilize, 21 of the 30 subjects reported that it did.

DISCUSSION

Hardware and Protocol: The goal to develop a sea state simulator to reproduce the movements and visual conditions that an astronaut will experience in the capsule after a water landing was achieved as part of this investigation. The use of equations to drive the pitch, roll, and heave movements of the Moog Motion Platform allowed for an easy way to vary the amplitude of the movement without affecting their underlying frequencies. Accommodating the natural tilt of floating capsules altered the original designs of the enclosure. Getting reliable information about the nominal tilt of the Orion capsule was difficult and the choice of a 10 deg static tilt was chosen before the information was available. At 10 deg it became necessary to remove the treadmill that normally sits atop the Moog. Doing so is undesirable because it adds significant time to hardware reconfiguration and setup, reducing the flexibility to accommodate simultaneous studies that require the motion platform. If the nominal tilts of the other capsules are near the 3 deg of the Orion, it would be possible to alter the enclosure design to rest on top of the treadmill for future studies.

With an eventual goal of using this protocol for the collection of a companion data set from returning astronaut crewmembers, efforts were made to minimize the time required to complete the protocol. The actual data collection time was consistently less than 15 minutes. The overall time was on the order of 45 minutes, but the majority of this time was used for familiarization and practice with the protocol. These could be accomplished in a single familiarization session and a shorter 15-20 minute test would be required for subsequent tests (i.e., postflight).

Test Results and Observations: The result that subjects take a longer time to complete the tasks when they are destabilized by movements of the support surface is not surprising. Subjective observations from the test sessions reveal a wide range in both the degree to which subjects were affected by the wave motion and the strategies that they used to combat it. Many stumbled during the execution of the tasks, but none fell. Subjects were instructed to avoid seeking support from the walls of the enclosure, but several were unable to avoid it. Others supported themselves by grabbing on the deployed ladder. Some used a strategy where they clearly delayed some of their own movements to wait for a period when the support surface motion was minimized at the top or bottom of a wave. Another source of variability in the results had more to do with the tasks themselves. Subjects were given the opportunity to practice the tasks until they felt comfortable with them but manipulating the carabiner-like clips and cam buckles to secure the nylon straps of the ladder, proved to be problematic at times. This is likely one source of the improvement that was observed in the STATIC condition based on test order.

Subjects were told that the line projected onto the walls during the HIGH w/ line condition was Earth-fixed and represented the horizon, but they were not exposed to it prior to the trial. Despite a lack of practice using it, 70% (21 of 30) of subjects reported that its presence was helpful. This improvement was not reflected in the data. Specific subject comments provide insight into why this might be the case. One subject who said the line was very helpful said, "I needed the line when I was on the floor." The nature of the tasks made it such that much of the time, while the subjects were doing activities near the floor, the line was not in their field of view. This can help explain the similarity in the task completion times between the HIGH and HIGH w/ line conditions. The kinematic parameters during the quiet stance portion of the high wave conditions also failed to show improvement. It is plausible that the measures that were derived were not sensitive enough to capture the subtle differences that the subjects perceived. Multiple subjects made a comment similar to, "The line allowed me to anticipate the movement before my body felt it." A more likely reason for the failure of the visual countermeasure to produce performance improvements is that it was not needed. Part of what the "body felt" was the vestibular response to the motion. Unlike returning astronauts, subjects in this study could rely on vestibular input. They still had a reliable representation of gravity available to them. They were not forced to use the visual countermeasure as the only source of reliable orientation information. This interpretation explains the order effect

	of the high wave conditions. Having experienced the conditions in the first trial, subjects were more able to use vestibular information in the second, resulting in improved stability. The learning effect observed from the data collected during the quiet stance portion of the trials could also be a change in strategy, particularly foot placement, between the first exposure and the second. Subjects unaware of the destabilizing effects of the test conditions often adopted a casual quiet stance with toes aligned parallel in the frontal plane and a more upright posture. Part way into the trial many changed positions, placing one foot ahead of the other and bending their knees and flexing at the waist to lower their center-of-mass. During the second exposure to the HIGH wave motion they immediately adopted the more cautious and stable stance. CONSIDERATIONS FOR FUTURE WORK Use of the protocol developed for this investigation would be suitable for consideration as part of a flight study to quantify postflight deficits and risks associated with water landings. As indicated, the time commitment was limited to allow it to fit in a crowded postflight timeline. Perhaps the most important factor revealed by this investigation is that such a task can be done safely. There is obviously a significant hardware consideration if there is a desire to do similar testing at a different location, but many of the time-consuming tasks (e.g., wave motion development and the enclosure design) would not have to be repeated. More work is required to determine the degree to which a visual countermeasure is effective for reducing instability after a water landing. The first goal of this investigation was to develop a test and collect normative data for later comparison to returning astronauts. As a result, the testing done used healthy subjects that could use vestibular input to provide a reliable gravitational reference. Therefore, the artificial horizon that was tested here did not provide the only veridical orientation information, as would be case for subjects with compromised vestibular input. To conduct a ground-based test to determine whether a visual orientation reference would be useful after flight, an effort needs to be made to disrupt the subjects' normal vestibular and sensorimotor function. It is also important to consider other presentations of the cues so that they could be visible during all portions of the activities being performed. NOTE (8/5/2020): Project continues with Dr. Brian Peters as new Principal Investigator, effective 7/27/2020; see that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Abstracts for Journals and Proceedings	Caldwell E, Peters B, Rosenberg M, Reschke M. "The effects of wave motion on capsule egress-like activities using ground-based subjects with and without a projected Earth-fixed visual horizon." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020